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Effects of water on the mechanical properties of silica glass using molecular dynamics

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ABSTRACT

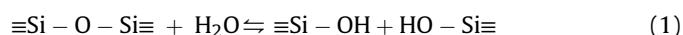
Understanding the effects of water on the mechanical properties of silica glass is important for many applications of silicate glasses. In this study, the effects of water on both elastic and plastic properties of pure silica glass are investigated. The introduction of molecular water leads to an increase of Young's modulus of silica glass at low water content, while hydroxyl exhibits an opposite effect. While hydroxyl groups decrease both strength and fracture toughness of the glass via destructing the silica network connectivity, molecular water undermines these properties by effectively driving the silica network to a "strained" configuration in absence of external stress. The water effect can be characterized in terms of the change of Si-O bond length and Si-O-Si bond angle. The plateau in the stress-strain curves of silica with the existence of molecular water in compact tension is associated with the Si-O bond breaking followed by formation of silanol groups. Moreover, the introduction of molecular water lowers the critical tensile stress where the plateau occurs.

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1. Introduction

Silicate glasses are broadly used in architecture [1], transportation [2], electronics [3], fiber optics [4] and information display [5] as either structural or functional materials. For any of these applications of silicate glass, water exhibits a huge impact on several critical glass properties, including mechanical [6–18], optical [19–21], electrical [22–24] and thermal properties [25–27]. Water can enter the silicate glass network during manufacturing as a component in the raw materials or penetrate into the glass while it is in service via diffusion or hydrolysis and condensation, e.g., from a humid environment [28]. The diffusion process depends on several factors, including temperature and pressure as shown in

experimental and theoretical studies [29–35]. In silicate glass, water exists as either molecules or hydroxyl groups. The reaction producing hydroxyl groups takes the following form [36]:



The ratio between molecular water and hydroxyl groups can vary with environment [21,37]. In the sodium silicate glass, water exists mainly as hydroxyls at small content or as molecules at large content [21]. As shown by Deubener et al. [38], even small amounts of water can have strong effect on glass properties such as the glass transition temperature. Mechanically, the crack initiation stress of silica glass immersed in water has been revealed to be time dependent, whereas it is time independent in non-aqueous liquids [15]; crack velocity is dependent on relative humidity in soda-lime glass [7]; fracture toughness of silica is reduced in aqueous environment [16]. The time dependence has been proposed to result from a stress-corrosion reaction consisting three steps: absorption of water to Si-O bond, concerted reaction and cleavage to form surface hydroxyls [36]. Water penetration has been found to lead to

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swelling of silica, which helps resist crack growth by developing a compressive stress zone to shield the crack tip [10,18], and accelerating surface structural relaxation [16,17]. Rimsza et al. [16] studied the effect of molecular water surrounding the crack-tip of silica and found that the fracture toughness is reduced due to water-silica interaction. However, the water can also exist in-situ within the silica network as either hydroxyl groups or molecules prior to the occurrence of a crack. While these roles of water are technically relevant, their effects are much less understood and are the central goal of this study.

Here, we introduce the two types of water (molecular and hydroxyl) directly into the silica network prior to mechanical tests and study their respective effects on the mechanical properties of the glass using molecular dynamics (MD) simulations. To the best of our knowledge, this is the first time that the impacts of the two species of water have been investigated separately. Although many experiments [6–9,12,14,15,17,21,26] have revealed the influence of water on glass, water molecules and hydroxyls have always been considered together rather than as separate species. Hence, there is a scientific need to isolate their respective roles in changing the mechanical properties of silica. Moreover, previous simulations have only considered the impact of water environment on the corrosion behavior of silica glass, where the glass is immersed in the aqueous environment. Unlike this study by Rimsza et al. [16], no further water is introduced during the following mechanical tests.

Our simulations show that, on the one hand, molecular water may increase the rigidity and decrease the compressibility of the silica glass network moderately; on the other hand, it weakens the network by driving it into a “strained” configuration without external stress. Through these two mechanisms, molecular water tends to decrease the fracture toughness of silica glass.

2. Simulation details

The in-situ water effects are investigated through MD simulations using LAMMPS [39]. The Yeon et al. ReaxFF [40] potential is used to simulate water in pure silica glass. We chose this particular ReaxFF Si/O/H parameter set since it was specifically developed to describe silica hydrolysis under mechanical load. ReaxFF has been demonstrated to be effective at reproducing the properties of silica

[41,42] and describes the reaction between silica and molecular water [43,44]. The structure of the glass is visualized with OVITO [45].

As shown in Fig. 1a, we constructed a bulk sample of the silica glass with a V-shape crack inside. The dimensions of the sample are initially $22.12 \text{ nm} (L_x) \times 22.46 \text{ nm} (L_y) \times 2.34 \text{ nm} (L_z)$. The sample was made by replicating a smaller sample of dimensions $3.69 \text{ nm} \times 3.74 \text{ nm} \times 2.34 \text{ nm}$ by $6 \times 6 \times 1$ in the three directions, which was obtained by melting a quartz at 4000 K for 500 ps, cooling the sample down to 300 K at a rate of 1 K/ps, and equilibrating the sample at 300 K for 100 ps. An integration timestep of 0.5 fs was used for the preparation of the pure silica glass, while 0.25 fs was used for all of the other simulations. The final density of the silica glass is 2.07 g/cm^3 . A Nosé-Hoover thermostat/barostat [46] in the isothermal-isobaric ensemble (NPT) was used to control the temperature of the system and maintain the pressure at 0 GPa.

The molecular water (unreacted water) and hydroxyl groups (pre-reacted water) were introduced into the silica network using different strategies. The former was introduced by randomly inserting water molecules into the voids in the silica network. The insertion was carried out such that the distance between the atoms in the water molecule and the network atoms is larger than 0.18 nm (larger than the sum of the radii of Si atom and O atom). The latter was introduced by randomly inserting water molecules into the glass as just described and then heating the sample at high temperature before cooling it down. In order to maintain the network structure as much as possible and highlight the effect of water, only the water molecules were heated at 2000 K (lower than T_g , T_g of the silica glass is above 2400 K based on the MD simulations when the cooling rate is 1 K/ps) while the glass network was maintained at 600 K using an NVT ensemble for 1 ns. During this time, a portion of the water molecules (>90%) react with the silica network via Eq. (1) and become hydroxyl groups. The atoms belonging to the original water molecules were then cooled down to 600 K with a cooling rate of 50 K/ps. Since the considered water content is limited ($\leq 18.5 \text{ mol}\%$) and the silica network is maintained at a lower temperature, the change of the silica network configuration is limited to bond break of -Si-O- and creation of hydroxyl groups. Here, taking sample 5 as an example, its network connectivity is partially broken due to the reaction between silica and molecular

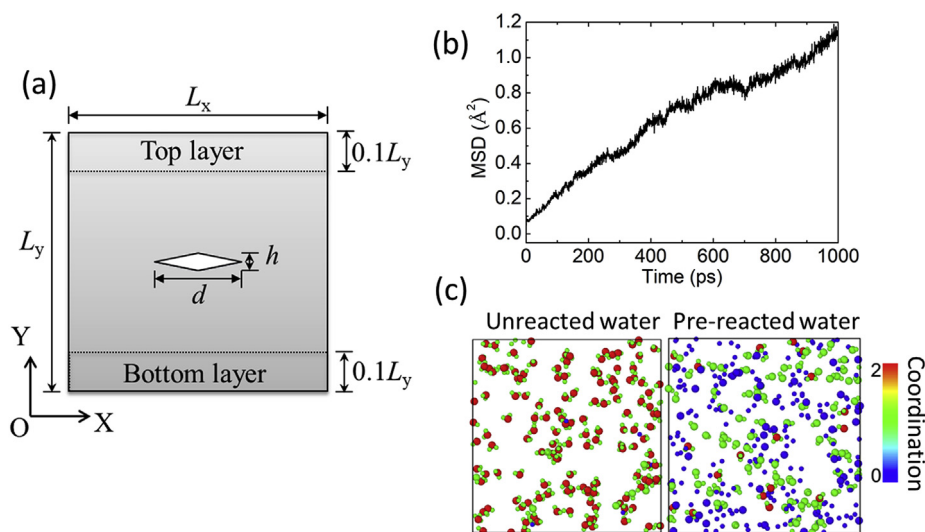


Fig. 1. (a) A sketch of the glass sample for compact tension test. The Z direction is perpendicular to the plane of the paper. (b) MSD of Si atoms in silica glass with 18.5 mol% pre-reacted water. (c) Water-related species distribution in a small sample ($\sim 3.7 \text{ nm} \times 3.7 \text{ nm} \times 2.3 \text{ nm}$) with 18.5 mol% unreacted water and 18.5 mol% pre-reacted water. The atoms in (c) are colored according to their coordination number; not all the atoms belonging to the original silica network are shown. The large circles denote oxygen atoms, while the small circles denote hydrogen atoms. All atoms come from the introduced water.

water, in contrast to sample 1. Mean square displacement (MSD) of silicon atoms in the sample 5 was given in Fig. 1b, where 0 ps corresponds to the moment when reaction between silica and water starts. The final MSD is $\sim 1.1 \text{ \AA}^2$ at 1000 ps, implying that the introduction of hydroxyl does not affect original silica configuration in a significant manner.

To create glass samples for compact tension tests, the atoms (including water) inside a V-shaped space were deleted as shown in Fig. 1a. To maintain charge equilibrium, the dimensions of the crack are slightly adjusted to ensure that the numbers of deleted Si (N_{Si}), O (N_{O}) and H (N_{H}) atoms follow $2N_{\text{Si}} + 0.5N_{\text{H}} = N_{\text{O}}$ (the samples with water) or $2N_{\text{Si}} = N_{\text{O}}$ (the samples without water). As shown in Fig. 1a, the initial crack length d is $\sim 5.4 \text{ nm}$ and aspect ratio h/d is ~ 0.25 among all the samples. To simplify the analysis and be similar to others [16,47–51], plane strain conditions are used in the X-Y plane and periodic boundary conditions are used in all directions. During the mechanical tests, the temperature of the system is maintained at 600 K ($T_{\text{g}} > 2400 \text{ K}$), a value at which the relaxation of the silica network can be ignored but the reaction between water and silica is facilitated due to the higher thermal energy compared to room temperature. Several samples were created in this work: a silica glass without water (sample 1); four silica glasses with unreacted molecular water: 5.4 mol%, 10.2 mol% (sample 2), 14.5 mol% and 18.5 mol% (sample 3); two silica glasses with pre-reacted water (hydroxyl): 10.2 mol% (sample 4) and 18.5 mol% (sample 5). The mole percent of water can be roughly converted to weight percent of water by dividing the former by 3 within the range of water content from 0 mol% to 18.5 mol%. The distribution of water (both molecular form and hydroxyl form) in two small samples without crack is illustrated in Fig. 1c.

To obtain Young's modulus and Poisson's ratio of various silica glasses, we performed a tension test using three small samples without cracks. They were obtained by melting and quenching following the same procedure as mentioned above. The dimensions of the samples are around $7.38 \text{ nm} \times 7.48 \text{ nm} \times 4.68 \text{ nm}$. A true strain rate of 10^9 s^{-1} was used. Linear fitting of true stress-strain relation within 1% strain was carried out to calculate the Young's modulus. The Poisson's ratio ν is given by σ_z/σ_y , while Young's modulus is given by $\sigma_y/\epsilon_y \cdot (1 - \nu^2)$.

To observe the brittle behavior of the glass without a crack, the sample size should be large enough according to Yuan et al. [52] and Shi et al. [48], and the minimal length can be calculated by $4\gamma_{\text{sf}}/E\epsilon_{\text{esl}}^2$, where γ_{sf} denotes surface energy and ϵ_{esl} elastic strain limit. For a glass with a crack, the brittle behavior can be observed if the sample length is longer than $l_{\text{min}} = \frac{4\gamma_{\text{sf}}}{E\epsilon_{\text{esl}}} + d$, where d denotes initial crack length. The surface energy is calculated by cutting a bulk glass sample into halves with two surfaces. We use the equation: $\gamma_{\text{sf}} = (E_{\text{s}} - E_{\text{B}})/A$, where E_{s} is the energy of system after cutting, E_{B} is the energy of the system prior to cutting and A is the surface area estimated to be twice of the cross-sectional area. ϵ_{esl} is obtained through a uniaxial-tension simulation. For sample 1, γ_{sf} is 1.37 J/m^2 and ϵ_{esl} is ~ 0.13 . The sample length used in the current study, $\sim 22 \text{ nm}$, is twice longer than l_{min} ($\sim 10.7 \text{ nm}$).

The fracture toughness K_{IC} of the glass was calculated using $K_{\text{IC}} = \sigma_y \sqrt{\frac{\pi d}{2}}$ [53]. σ_y is critical far-field stress for the initiation of crack propagation. The plastic zone size r_p was estimated by $r_p = \frac{1}{6\pi} \left(\frac{K_{\text{IC}}}{\sigma_{\text{ys}}} \right)^2$ [53]. The yield strength σ_{ys} was obtained using a small sample without a crack ($3.69 \text{ nm} \times 3.74 \text{ nm} \times 2.34 \text{ nm}$) and ranges from 10.6 to 15.7 GPa. The obtained r_p ($\sim 0.2 \text{ nm}$) is much smaller than the crack sizes $d/2$ (larger than 2.0 nm) and sample sizes. Therefore small scale yielding assumption is satisfied and linear

elastic fracture mechanics can be used.

3. Results and discussion

3.1. Water effects on elastic properties

Fig. 2 shows the evolution of Young's modulus, density, bulk modulus, and Poisson's ratio with water content. The obtained Young's modulus and shear modulus for pure silica are 60 GPa and 23 GPa, comparable to experimental results of 73 GPa and 31 GPa (at 300 K). In Fig. 2a, the introduction of hydroxyl leads to the decrease in coordination number of Si atoms bonded to bridging O atoms due to Si-O bond breaking, which corresponds to a loss of connectivity of the silica network, and therefore a decrease of Young's modulus is expected. This is consistent with the results on silica glasses by Parc et al. [14] where they showed that the longitudinal modulus becomes smaller at a higher concentration of hydroxyls. More interestingly, the introduction of unreacted molecular water has led to a moderate increase of Young's modulus, even though the silica network is swelling after the introduction of molecular water as can be seen from the density change of silica glass in Fig. 2b. This increase in Young's modulus may come from the steric hindrance of water molecules when they are inserted into the silica network. Experimentally, Ito et al. [54] have observed that Young's modulus of sodium silicate glasses decreases gradually with increasing water content. Based on our simulation results and the work of Acocella et al. [21], it is reasonable to conjecture that water within their samples might exist mainly in the form of hydroxyl groups, especially at low content ($< \sim 18 \text{ mol\%}$).

Fig. 2b shows the change of the glass density as a function of the water content. The swelling of the glass due to both types of water can be clearly seen from the decrease in the density of the silica network (ρ_{SiO_2}). By comparison, the swelling due to hydroxyl is larger than that due to molecular water because the latter does not change the network connectivity. Experiments have revealed a linear dependence of volume swelling on water concentration [18,55–58]. Given that the silica network is not significantly changed during preparation, the volume swelling strain is $\sim 7\%$ for sample 5 (18.5 mol% unreacted water), which is close to $11\%–12.5\%$ obtained according to experimental findings.

The bulk modulus (see Fig. 2c) increases with the introduction of molecular water, indicating that the compressibility of silica network becomes lower when it is stuffed with water molecules. In contrast, as a result of the reduction in the silica network connectivity when hydroxyl groups are introduced, the bulk modulus decreases moderately. However, the two types of water will lead to a similar increase in Poisson's ratio (see Fig. 2d), suggesting that water in either form tends to facilitate shear deformation in the glass.

3.2. Water effects on fracture behaviors

The stress-strain curves from the compact tension test of samples 1–5 are shown in Fig. 3a. For pure silica glass, the tensile stress reaches a peak value ($\sim 10 \text{ GPa}$) beyond which it drops abruptly down to zero. In contrast, a plateau of the tensile stress exists in all the other water-containing samples. The shear strain distributions of sample 1 and 3 at the peak values (as indicated by the arrows in Fig. 3a) are given in Fig. 3b, d and 3e. The plateaus are characterized by the enlargement of red area around the crack tip (from Fig. 3d at strain = 11.9% to Fig. 3e at strain = 13.8%) wherein the shear strain is ≥ 0.4 , relating to the crack growth in the silica glass with molecular water prior to the final fracture of silica glass. This can be directly observed from the atomic configuration of sample 3 in Fig. 4 at the two values of strain. The crack growth proceeds via the

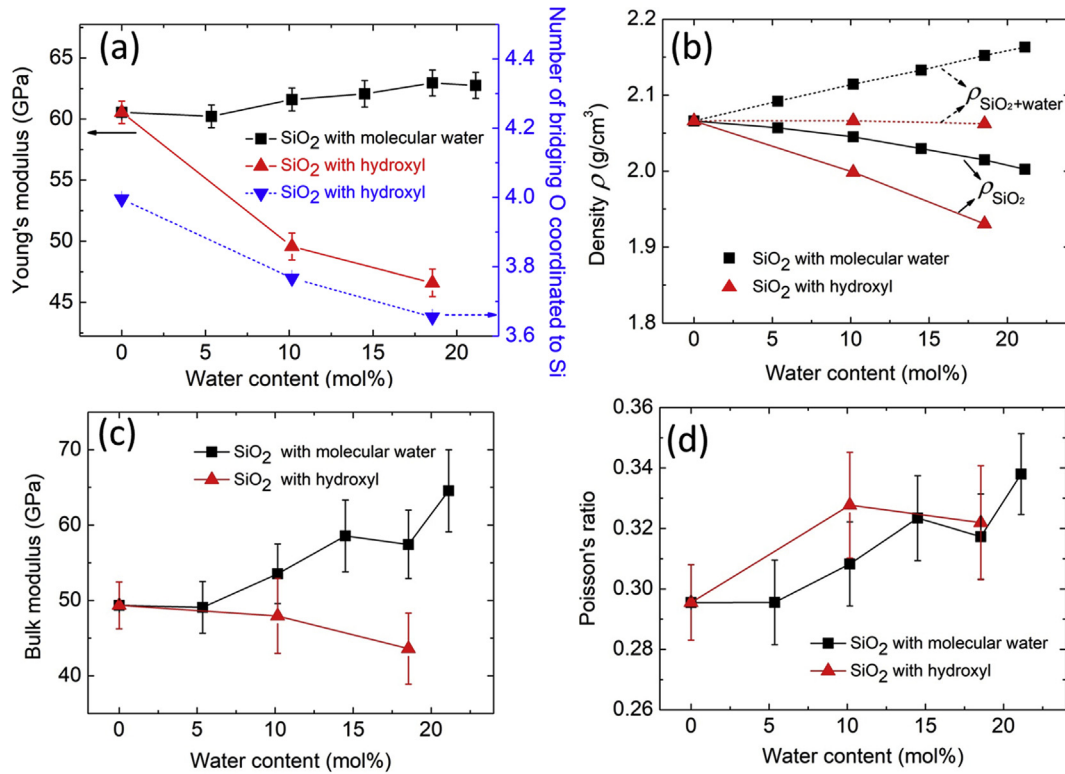


Fig. 2. (a) Young's modulus, (b) density, (c) bulk modulus and (d) Poisson's ratio of silica before and after the introduction of water. In (b), solid lines denote densities without water, while dashed lines densities with water.

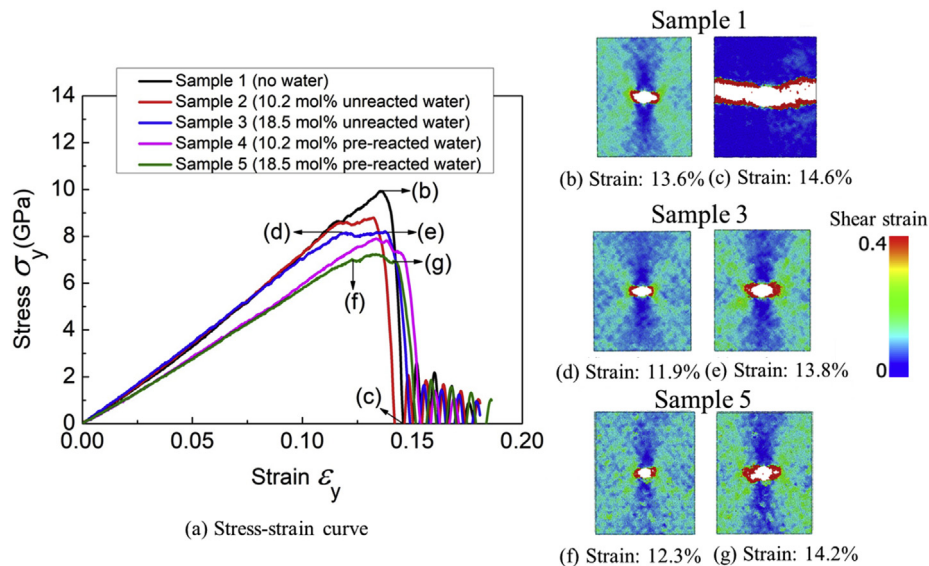


Fig. 3. (a) Stress-strain curves using true strain and stress; (b–g) shear strain distributions at the specified strains close to the peaks in the (a). Atoms are colored according to the von Mises shear strain from 0 (blue) to 0.4 (red). Water-related species are not shown in (b–g). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

dissociation of Si-O bonds (see the black bonds denoted in Fig. 4a) during the loading process, which partially releases the elastic energy and keeps the stress from further escalation. After the breaking of Si-O bonds, the freshly formed crack surface is stabilized by formation of silanols at the crack surface (as shown in Fig. 4c), which lowers the surface energy of the crack according to Rimsza et al. [59]. These can also be used to explain the plateaus in

the samples with pre-reacted water, since not all of the introduced molecular water reacts with silica and a small amount of molecules (around 10% of pre-reacted water) is left. The plateau in the stress-strain curves from the MD results is consistent with the time-dependent nature of the fracture behavior of silica glass in the presence of either types of water as reported in the experiments [7,15]. Conversely, if the applied strain is fixed once the tensile

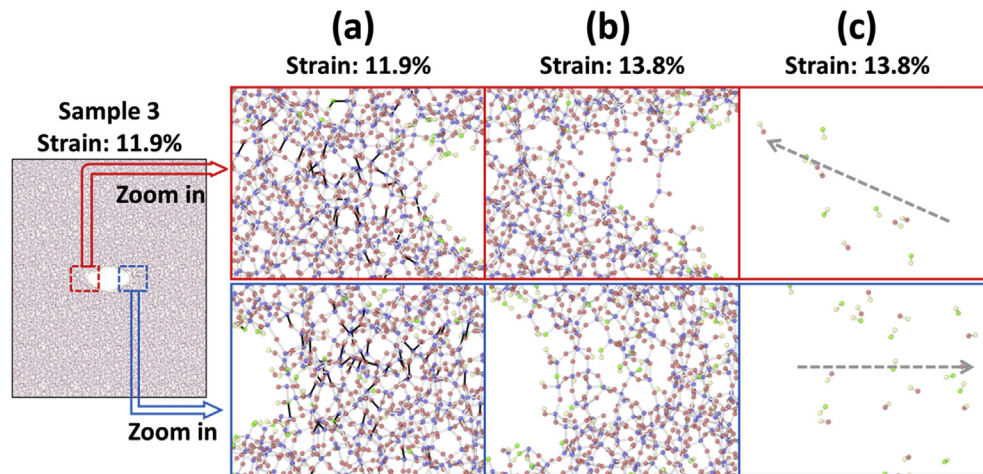


Fig. 4. Atomic configuration of silica glass with 18.5 mol% molecular water (sample 3) at the initial stage of crack propagation. Strains 11.9% (a) and 13.8% (b–c) correspond to the start and end points of plateau in the stress-strain curve respectively. For clarification, molecular water is not shown. Different colors of balls denote different types of atoms: blue – silicon atom; red – oxygen atom come from original silica network; green – oxygen atom come from original molecular water; yellow – hydrogen atom. Black bonds in (a) denote the broken bonds during tension from strain 11.9%–13.8%, while the atoms in (c) indicate the newly formed hydroxyl. The dashed arrows in (c) are guides to show the propagation direction of the crack. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

stress reaches the plateau, the stress will decrease with time due to continuous Si-O bond breaking. To test this phenomenon, we extracted the configuration of sample 3 corresponding to a strain of 11.9% as shown in Fig. 3d and then fixed the strain along the loading direction for 150 ps. The crack continued to grow in the direction perpendicular to the loading direction and the stress decreased as seen in Fig. 5. Based on the previous analysis, the starting point of the plateau can be treated as the crack propagation moment of the material, and accordingly the stress at the plateau is used as the critical tensile stress (σ'_y) at a strain rate of 10^9 s^{-1} . The critical tensile stress decreases from sample 1 to sample 5, as seen in Fig. 3a; therefore, introduction of both molecular and hydroxyl water leads to a lower strength of the silica glass.

The plateau observed in the stress-strain curves can be further understood from the change of Si-O bond length and Si-O-Si bond angle as plotted in Fig. 6. With the introduction of both molecular water and hydroxyl groups, the change of silica configuration at 0% strain is shown in Fig. 6a and b. It can be seen that molecular water has led to extension of the Si-O bond length and opening of the Si-O-Si bond angle in samples 2 and 3. In the strained silica glass without water, extension of bond length and opening of bond angle at crack tip were found by Rimsza et al. [60]. IR spectroscopy also showed that the Si-O-Si bond angle increased under tensile stress [61]. Therefore, according to our simulation, introduction of molecular water has effectively driven the silica network to a “strained” configuration without mechanical tension. Consequently, less external stress is needed for the breaking of Si-O bonds and crack propagation. The introduction of hydroxyl (with small

amount of unreacted molecular water left in samples 4 and 5) not only changes the bond length and bond angle, but also weakens the connectivity of the silica network (discussed in Section 3.1). Hence, the lowering of strength in samples 4 and 5 is larger compared to that in samples 2 and 3. The change of Si-O bond length and Si-O-Si bond angle distribution of sample 3 before and after the plateau are given in Fig. 6c and d. The Si-O bond length has extended prior to the plateau and remains relatively constant during the plateau, in contrast, the Si-O-Si bond angle is more open prior to the plateau and becomes less so after the plateau. The variations of bond length and bond angle during the plateaus indicate that the silica glass has not been further strained.

3.3. Molecular water effect on near-crack stress

As shown above, the crack propagation occurs when the stress around the crack tip, i.e., the near-crack stress ($\sigma_{\text{near-crack}}$) reaches a critical value, $\sigma'_{\text{near-crack}}$. The local stress profiles along the X direction at $y = 0.5L_y$ (solid curves) and $y = 0.05L_y$ (dashed curves) are plotted in Fig. 7a. Except for the left-most points of the solid curves, the stress profiles are close to each other among all samples, suggesting that the connectivity of the original silica network is hardly affected. We note water-related species were not considered in the stress analysis. For convenience, the average stress at a location close to the crack tip ($d_{\text{crack}} = 0-0.8 \text{ nm}$, see the inset of Fig. 7a), i.e., the left-most points of the solid lines in Fig. 7a, is used as the $\sigma_{\text{near-crack}}$. The stress far from the crack, i.e., the far-field stress $\sigma_{\text{far-field}}$, is obtained by averaging the stress at the top and bottom layers of the samples, see Fig. 1a. Because the crack-tip shape is better defined prior to the crack propagation, we have estimated the relation between $\sigma_{\text{near-crack}}$ and $\sigma_{\text{far-field}}$ when the strain is $\sim 11.5\%$ (a value that is close to but lower than the critical strain when the crack starts to propagate) as shown in Fig. 7b. According to Inglis [62], $\sigma_{\text{near-crack}}/\sigma_{\text{far-field}}$ is related to the shape of the crack. Because the change of crack shape from the strain 11.5% to the beginning of crack propagation is little, Fig. 7b effectively shows that the $\sigma_{\text{near-crack}}/\sigma_{\text{far-field}}$ decreases with water content when crack is about to propagate. The inset of Fig. 7b shows that the far-field stress and tensile stress σ_y are close to each other with a ratio of ~ 1.01 . Assuming the ratio of the critical near-crack stress ($\sigma'_{\text{near-crack}}$) to

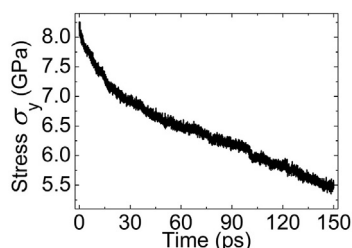


Fig. 5. Stress evolution with time when the strain is fixed at 11.9% in sample 3.

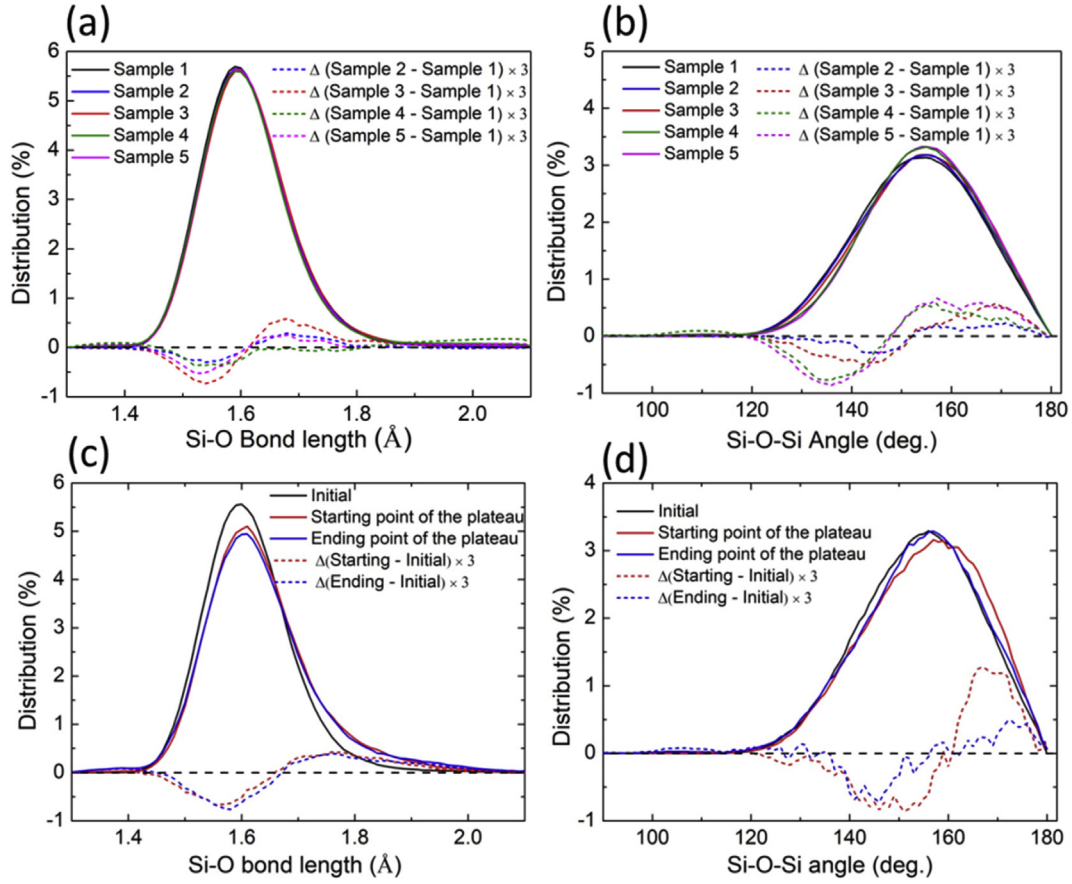


Fig. 6. Distribution of Si-O bond length and Si-O-Si bond angle. (a) Si-O bond length distribution of all the samples at 0% strain; (b) Si-O-Si bond angle distribution of all the samples at 0% strain; (c) Si-O bond length distribution around crack tips of the sample 3; (d) Si-O-Si bond angle distribution around the crack tips of the sample 3.

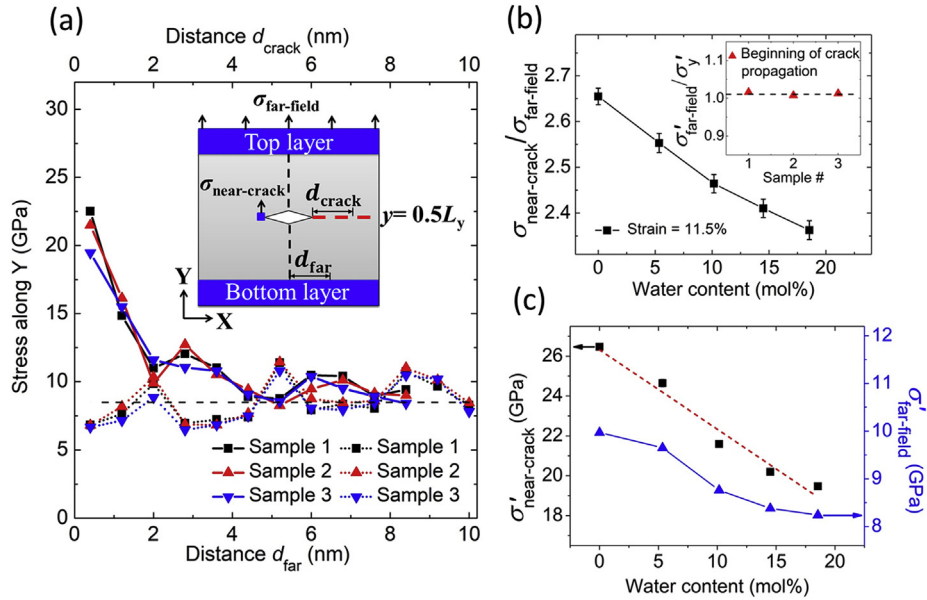


Fig. 7. (a) Stress distribution at a strain of 11.5% along $y = 0.5L_y$ (solid curves) and $y = 0.05 L_y$ (dashed curves) with $y = 0.5L_y$ being indicated by a red dashed line; (b) $\sigma_{near-crack}/\sigma_{far-field}$ (near-crack stress/far-field stress) at strain = 11.5% at different water contents; the inset shows the ratio of $\sigma'_{far-field}/\sigma'_y$ at the beginning of crack propagation; (c) $\sigma'_{near-crack}$ and $\sigma'_{far-field}$ at the beginning of crack propagation with different molecular water contents. The red linear-fitting curve in (c) has a correlation coefficient of 0.98. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the critical far-field stress ($\sigma'_{\text{far-field}}$) is the same as $\sigma_{\text{near-crack}}/\sigma_{\text{far-field}}$, i.e., $\sigma'_{\text{near-crack}}/\sigma'_{\text{far-field}} = \sigma_{\text{near-crack}}/\sigma_{\text{far-field}}$. $\sigma'_{\text{near-crack}}$ can then be calculated from the critical tensile stress ($\sigma'_{\text{far-field}} \approx 1.01\sigma_y$, see Section 3.2) and is found to decrease linearly with water content, as shown in Fig. 7c. The reduction of critical stress due to water can be understood from the bond-breaking of a Si-O bond as schematically shown in Fig. 8. For pure silica, the energy barrier overcome by exerting external stress to break a Si-O bond is denoted as E_{stress} . With the introduction of molecular water, the energy barrier decreases to E'_{stress} because the silica network has already been driven to a “strained” configuration as demonstrated in Section 3.2 and Fig. 6. This is consistent with the work by Potter et al. [25], which theoretically revealed that molecular water tends to weaken the constraint associated with Si-O-Si bond angle and correspondingly decrease the energy required to results in a rigid-to-floppy transition of the constraint. The inset of Fig. 8 shows a selected instantaneous configuration in one of the glass samples with molecular water when a hydrogen bond ($\sim 2.1 \text{ \AA}$) forms between a water molecule and a bridging O atom, through which the silica network can be strained by the water in the absence of external stress.

3.4. Water effects on fracture toughness

The fracture toughness of the five glass samples are given in Fig. 9. Without water, the fracture toughness of pure silica glass (sample 1, $\sim 0.92 \text{ MPa m}^{1/2}$) is close to the experimental value ($0.6\text{--}0.8 \text{ MPa m}^{1/2}$) [63–66]. The overestimation of fracture toughness could be due to the inaccurate critical tensile stress from using a high loading rate of 10^9 s^{-1} . Introduction of molecular water leads to a decrease of the fracture toughness of silica glass and an even larger decrease in the case of hydroxyl. The decreases in fracture toughness for samples 2–5 are 18.5%, 24.4%, 26.0% and 39.5%, respectively, which are consistent with experimental results [63–66] showing that the average fracture toughness of wet silica (i.e. silica soaked in aqueous solution or water vapor) is 22.1% lower than that of dry silica. While either type of water can exist in the silicate glass, our results demonstrate that both can effectively

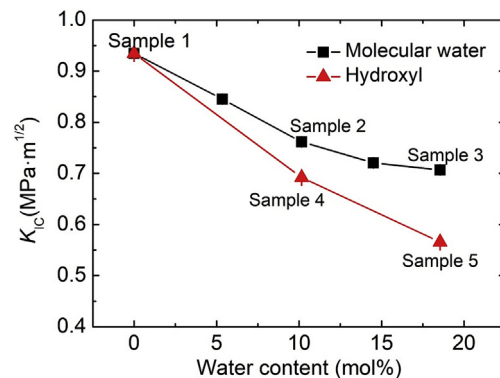


Fig. 9. Fracture toughness of silica glasses at 600 K. The error bars are smaller than the symbols.

decrease the fracture toughness through either straining the silica network or breaking its connectivity as demonstrated in Sections 3.1 and 3.2. We note that the decrease of fracture toughness due to water is *not* inconsistent with the theoretical results by Wiederhorn et al. [10,18] where the water is only introduced to the crack tip and causes a *localized* swelling zone which helps resist crack growth. In contrast, the water introduced to the whole silica glass sample in the current study is randomly placed in the network and, therefore, the swelling is relatively homogeneous. To this extent, our simulations may correspond to a long-term behavior of the silica glass in aqueous environment so that water molecules may diffuse further into the glass network and any local swelling effect becomes less important.

Water-glass interactions and the degradation of glasses in aqueous solutions have attracted a great deal of interest for various applications of inorganic glasses [7,10–13,18,29,30,36,66–68]. Our MD simulation results on the effects of individual water species can be very useful for understanding the impact of glass corrosion on its mechanical behavior. For example, the current methods can be combined with kinetic Monte Carlo simulations [69–71] to predict the evolution of glass mechanical properties over long time scales, which is of great interest for compositional design of advanced glasses. Isolating the effects of the two forms of water can be also used as input for finite element models, e.g., in the study of the cracking behavior of silicate glasses in a humid environment.

4. Conclusion

To conclude, a better understanding of the effects of water in molecular or hydroxyl forms on both elastic and plastic properties of pure silica is achieved via using ReaxFF based reactive molecular dynamics simulations. While introduction of molecular water into the silicate network leads to an increase in both Young's modulus and bulk modulus of silica glass, hydroxyl groups exhibit an opposite effect on both quantities by destructing the connectivity of the network. The increase in Poisson's ratio of silica glass in the presence of either type of water suggests that water tends to facilitate shear deformation. In compact tension tests of the silica glass, the stress-strain curves of water-containing samples feature a plateau during which the crack propagates via Si-O bond break and formation of silanol groups. The critical stress at which this plateau occurs decreases with the molecular water content, which is understood in terms of a “strained” silica network in existence of water and in absence of external stress. Such effect can be characterized by the extension of Si-O bond length and the opening of Si-O-Si bond angle found in silica glass with water. Both molecular and hydroxyl types of water are shown to decrease the strength and

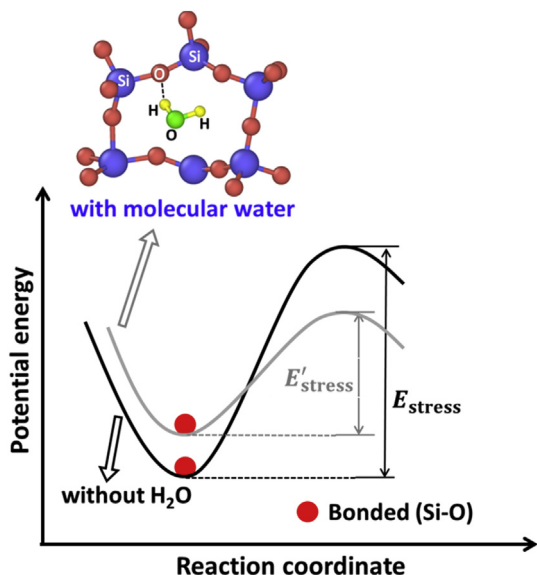


Fig. 8. Sketch of potential energy vs. reaction coordinate for Si-O bond break without water (black line) and with molecular water (gray line). E_{stress} and E'_{stress} denote the energy barrier to be overcome without and with molecular water, respectively. The difference between E_{stress} and E'_{stress} denotes the decrease of energy barrier due to molecular water. The atomic configuration on the top illustrates the effect of a water molecule on the glass through a hydrogen bond, which may strain the glass network.

fracture toughness of silica by either straining the silica network or destructing its connectivity. With experimental glass samples often containing a mixture of both types of water, the current simulation results provide us with a better understanding of their respective effects on silica glass. Therefore, our study can be useful in, e.g., building of mechanical model of water-containing silicate glass or prediction of the silicate glass mechanical properties either at different proportion of the two types of water or after long-term corrosion in aqueous solution.

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